



UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

STATISTICAL SIGNAL PROCESSING - *Prova scritta del 18/07/2024*

EXERCISE

It is given a Moving Average (MA) process $S[n]$ of order 1, $S[n] = I[n] - 0.9 \cdot I[n-1]$, where $I[n]$ is the zero-mean innovation process with variance $\sigma_I^2 = 1$.

(1) Derive the system function $L(z)$ of the innovation filter of the process $S[n]$, its zeros and poles, and determine if it is a low-pass, high-pass, or band-pass filter. Derive the Autocorrelation Function (ACF) $R_S[m] = E\{S[n]S[n+m]\}$ and the Power Spectral Density (PSD) $S_S(e^{j2\pi f}) = FT\{R_S[m]\}$ of the process $S[n]$ and plot them.

(2) By applying the Orthogonality Principle, derive the optimal linear predictor of order $P=2$ of the signal process $S[n]$, i.e. the Linear Minimum Mean Square error (LMMSE) estimator of $S[n]$ given the observed data $S[n-1]$ and $S[n-2]$; derive the bias $b \triangleq E\{\varepsilon[n]\} = E\{S[n] - \hat{S}[n]\}$ and the Mean Square Error (MSE) $\sigma_\varepsilon^2 \triangleq E\{\varepsilon^2[n]\} = E\{(S[n] - \hat{S}[n])^2\}$.

(3) Assuming that the signal of interest $S[n]$ is observed in the presence of additive white noise, i.e. $X[n] = S[n] + W[n]$, where $W[n]$ is additive white noise with variance $\sigma_W^2 = 2$, by applying the Orthogonality Principle again, derive the optimal linear predictor of order $P=2$ of the process $S[n]$, i.e. the LMMSE estimator of $S[n]$ given the observed data $X[n-1]$ and $X[n-2]$; derive its MSE and compare it with the one previously derived.

(4) If we know that the signal process $S[n]$ and the observed process $X[n]$ are jointly Gaussian, what can we additionally say about the linear predictor $\hat{S}[n]$?



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Solution.

(1) Process MA(1): $S[n] = I[n] - 0.9I[n-1]$

$$S(z) = I(z) - 0.9z^{-1}I(z)$$

$$L(z) = \frac{S(z)}{I(z)} = 1 - 0.9z^{-1} = \frac{z - 0.9}{z}$$

The innovation filter has 1 zero at $z = 0.9$ and 1 pole at $z = 0$. Consequently, the innovation filter is a high-pass filter, and so the process is a high-pass process.

The ACF and the PSD are given by:

$$\begin{aligned} R_s[m] &= E\{S[n]S[n+m]\} = E\{(I[n] - 0.9 \cdot I[n-1])(I[n+m] - 0.9 \cdot I[n+m-1])\} \\ &= R_I[m] - 0.9 \cdot R_I[m-1] - 0.9 \cdot R_I[m+1] + 0.81 \cdot R_I[m] \\ &= 1.81 \cdot \sigma_I^2 \delta[m] - 0.9 \cdot \sigma_I^2 \delta[m-1] - 0.9 \cdot \sigma_I^2 \delta[m+1] \\ &= 1.81 \cdot \delta[m] - 0.9 \cdot \delta[m-1] - 0.9 \cdot \delta[m+1] \end{aligned}$$

$$R_s[m] = \begin{cases} 1.81, & m = 0 \\ -0.9, & m = \pm 1 \\ 0, & |m| \geq 2 \end{cases}$$

$$\begin{aligned} S_s(e^{j2\pi f}) &= FT\{R_s[m]\} = \sum_{m=-\infty}^{+\infty} R_s[m]e^{-j2\pi fm} = \sum_{m=-\infty}^{+\infty} [1.81 \cdot \delta[m] - 0.9 \cdot \delta[m-1] - 0.9 \cdot \delta[m+1]]e^{-j2\pi fm} \\ &= 1.81 - 0.9e^{-j2\pi f} - 0.9e^{j2\pi f} = 1.81 - 1.8\cos(2\pi f) \end{aligned}$$

$$S_s(e^{j2\pi f}) = 1.81 - 1.8\cos(2\pi f)$$

$S[n]$ is a high-pass process.

$$(2) \hat{S}[n] = h[1]S[n-1] + h[2]S[n-2] + b$$

Since all the processes are zero mean we have that: $E\{\varepsilon[n]\} = 0 \Rightarrow b = 0$.

$$E\{\varepsilon[n]S[n-m]\} = 0, \quad m = 1, 2$$

$$E\{(S[n] - \hat{S}[n])S[n-m]\} = 0, \quad m = 1, 2$$

$$E\{(S[n] - h[1]S[n-1] - h[2]S[n-2])S[n-m]\} = 0, \quad m = 1, 2$$

$$R_s[m] - h[1]R_s[m-1] - h[2]R_s[m-2] = 0, \quad m = 1, 2$$



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$$\begin{cases} h[1]R_s[0] + h[2]R_s[-1] = R_s[1] \\ h[1]R_s[1] + h[2]R_s[0] = R_s[2] \end{cases}$$

$$\begin{bmatrix} R_s[0] & R_s[-1] \\ R_s[1] & R_s[0] \end{bmatrix} \begin{bmatrix} h[1] \\ h[2] \end{bmatrix} = \begin{bmatrix} R_s[1] \\ R_s[2] \end{bmatrix}$$

$$\begin{bmatrix} 1.81 & -0.9 \\ -0.9 & 1.81 \end{bmatrix} \begin{bmatrix} h[1] \\ h[2] \end{bmatrix} = \begin{bmatrix} -0.9 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} h[1] \\ h[2] \end{bmatrix} = \begin{bmatrix} -0.6606 \\ -0.3285 \end{bmatrix}$$

$$b \triangleq E\{\varepsilon[n]\} = E\{S[n] - \hat{S}[n]\} = E\{S[n] - h[1]E\{S[n-1]\} - h[2]E\{S[n-2]\}\} = 0$$

The estimator is unbiased.

$$\begin{aligned} \sigma_\varepsilon^2 &= E\{\varepsilon^2[n]\} = E\{(S[n] - \hat{S}[n])S[n]\} = E\{(S[n] - h[1]S[n-1] - h[2]S[n-2])S[n]\} \\ &= R_s[0] - h[1]R_s[1] - h[2]R_s[2] = 1.81 - (-0.6606) \cdot (-0.9) - (-0.3285) \cdot 0 = 1.2155 \end{aligned}$$

$$\sigma_\varepsilon^2 = 1.2155$$

$$(3) \quad \hat{S}[n] = h[1]X[n-1] + h[2]X[n-2] + b$$

Since all the processes are zero mean we have again that $E\{\varepsilon[n]\} = 0 \Rightarrow b = 0$.

$$E\{\varepsilon[n]X[n-m]\} = 0, \quad m = 1, 2$$

$$E\{(S[n] - \hat{S}[n])X[n-m]\} = 0, \quad m = 1, 2$$

$$E\{(S[n] - h[1]X[n-1] - h[2]X[n-2])X[n-m]\} = 0, \quad m = 1, 2$$

$$R_{sx}[-m] - h[1]R_x[m-1] - h[2]R_x[m-2] = 0, \quad m = 1, 2$$



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$$\begin{cases} h[1]R_x[0] + h[2]R_x[-1] = R_{sx}[-1] = R_s[1] \\ h[1]R_x[1] + h[2]R_x[0] = R_{sx}[-2] = R_s[2] \end{cases}$$

$$\begin{bmatrix} R_x[0] & R_x[-1] \\ R_x[1] & R_x[0] \end{bmatrix} \begin{bmatrix} h[1] \\ h[2] \end{bmatrix} = \begin{bmatrix} R_s[1] \\ R_s[2] \end{bmatrix}$$

$$R_x[m] = R_s[m] + R_w[m] = R_s[m] + \sigma_w^2 \delta[m] = R_s[m] + 2\delta[m]$$

$$\begin{bmatrix} 3.81 & -0.9 \\ -0.9 & 3.81 \end{bmatrix} \begin{bmatrix} h[1] \\ h[2] \end{bmatrix} = \begin{bmatrix} -0.9 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} h[1] \\ h[2] \end{bmatrix} = \begin{bmatrix} -0.2502 \\ -0.0591 \end{bmatrix}$$

$$\begin{aligned} \sigma_\varepsilon^2 &= E\{\varepsilon^2[n]\} = E\{(S[n] - \hat{S}[n])S[n]\} = E\{(S[n] - h[1]X[n-1] - h[2]X[n-2])S[n]\} \\ &= R_s[0] - h[1]R_s[1] - h[2]R_s[2] = 1.81 - (-0.2502) \cdot (-0.9) - (-0.0591) \cdot 0 = 1.5848 \end{aligned}$$

$$\sigma_\varepsilon^2 = 1.5848$$

which is obviously greater than the MSE in the previous case, because of the presence of additive noise $W[n]$.

(4) If we know that the signal process $S[n]$ and the observed process $X[n]$ are jointly Gaussian, we can say that: (1) the linear predictor $\hat{S}[n]$ is Gaussian distributed; (2) $\hat{S}[n]$ is not only the LMMSE estimator but it also the MMSE estimator, i.e. there is no non-linear estimator with better performance; (3) in the Gaussian case this estimator is efficient, i.e. its MSE coincide with the Bayesian Cramér-Rao bound (BCRB).